

EIGRP - Enhanced Interior Gateway Routing Protocol

Enhanced Interior Gateway Routing Protocol (EIGRP) is an advanced distance-vector routing protocol developed by Cisco. It combines the best features of distance-vector and link-state protocols to provide fast convergence, efficient use of network resources, and scalability. Here's a detailed overview of EIGRP, its features, operations, and configuration examples.

Key Features of EIGRP

1. Hybrid Protocol:

EIGRP is often called a hybrid protocol because it incorporates characteristics of both distance-vector and link-state protocols.

2. Fast Convergence:

EIGRP uses the Diffusing Update Algorithm (DUAL) to achieve fast convergence and ensure loop-free routes.

3. Efficient Use of Bandwidth:

EIGRP only sends incremental updates, reducing the amount of bandwidth used for routing updates.

4. Scalability:

EIGRP can support large and complex networks due to its efficient use of network resources and fast convergence.

5. Support for VLSM and CIDR:

EIGRP supports Variable Length Subnet Masking (VLSM) and Classless Inter-Domain Routing (CIDR), allowing for more flexible network design.

6. Unequal-Cost Load Balancing:

EIGRP supports unequal-cost load balancing, allowing traffic to be distributed over multiple paths with different metrics.

7. Authentication

EIGRP supports MD5 authentication to ensure that routing updates are authentic.

8.Protocol-Independent:

EIGRP can route multiple Layer 3 protocols, such as IPv4, IPv6, IPX, and AppleTalk.

Some Important points about EIGRP:

- EIGRP AD VALUE:

- . Internal AD Value: 90
- . External AD Value: 170

- EIGRP Routing Code:

- . D: Internal
- . D*Ex: External

- Maximum Hop Count in EIGRP 100.

- Maximum path: 4

- Multicast IP Address of EIGRP: 224.0.0.10

- Algorithm for best Path (DUAL): Diffusion Update Algorithm

- **EIGRP States:**

- . Pending
- . Full State

- **EIGRP is a Reliable Transfer Protocol**

- **EIGRP SIA Mode: Stuck Active Mode 180 Second**

The SIA Timer helps detect when a route is stuck in an active state due to a lack of response from neighbors.

- **EIGRP Supports classless routing protocol**

- **Protocol Number 88**

- **Supports IP, Ipx, Apple talk**

- **EIGRP has concept of AS number Range (1-65536)**

Timers in EIGRP

Hello Timer

The Hello Timer determines how often EIGRP sends Hello packets to discover and maintain neighbor relationships.

Default Values:

High-speed interfaces (e.g., Ethernet): 5 seconds. Low-speed interfaces (e.g., WAN links): 60 seconds.

Hold Timer:

The Hold Timer specifies how long a router should wait for a Hello packet from a neighbor before declaring the neighbor unreachable.

Default Values:

High-speed interfaces: 15 seconds (three times the Hello interval).

Low-speed interfaces: 180 seconds.

EIGRP Packet Types

EIGRP uses several packet types for its operations:

Hello Packets: Used to discover and maintain neighbor relationships.

Update Packets: Used to propagate routing information.

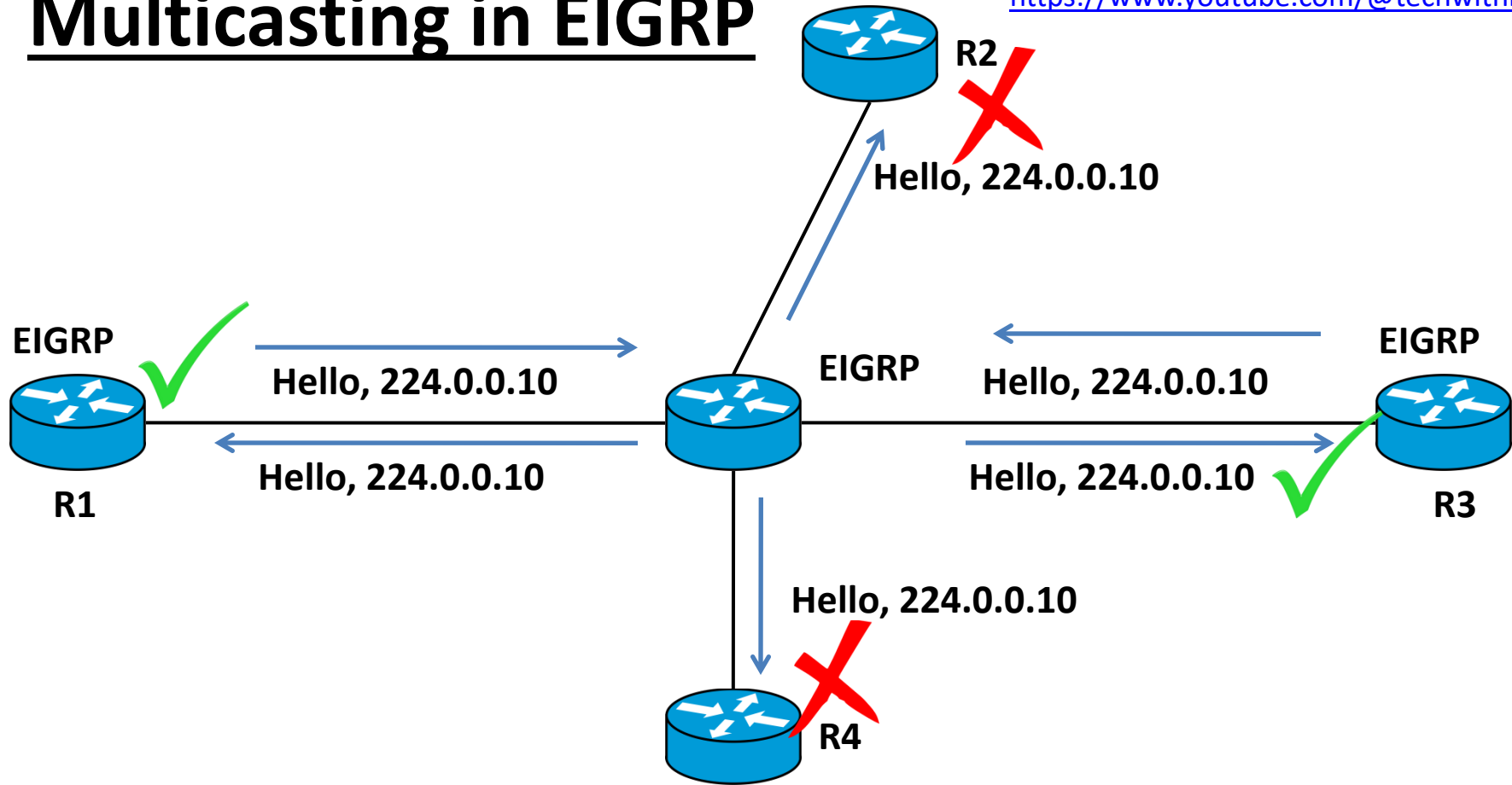
Query Packets: Used to request routing information from neighbors.

Reply Packets: Used to respond to query packets.

Acknowledgment Packets: Used to acknowledge the receipt of update, query, and reply packets.

Multicasting in EIGRP

<https://www.youtube.com/@techwithravish>



The Hello packet will be exchanged in between those routers running EIGRP

Tables in EIGRP

Neighbor Table: Tracks all EIGRP neighbors, including their status and reachability.

Router# show ip eigrp neighbors

Topology Table: Stores all routes advertised by neighbors and is used by DUAL to calculate the best paths.

Router# show ip eigrp topology

Routing Table: Contains the best paths to each destination, as determined by EIGRP, and is used for packet forwarding.

Router# show ip route

EIGRP Metrics:

EIGRP uses a composite metric based on five parameters:

Bandwidth: The minimum bandwidth of the route.

Delay: The cumulative delay of the route.

Load: The load on the route (optional).

Reliability: The reliability of the route (optional).

MTU: The Maximum Transmission Unit (not used in the metric calculation).

The default EIGRP metric formula primarily uses bandwidth and delay:

$$\text{Metric} = ((10^7 / \text{Minimum Bandwidth}) + \text{Cumulative Delay}) \times 256$$

EIGRP Neighbor Discovery and Maintenance

1. Hello Packets:

- Sent periodically to discover and maintain neighbor relationships.
- Default hello interval is 5 seconds on high-speed interfaces and 60 seconds on low-speed interfaces.

2. Neighbor Table:

- EIGRP maintains a table of all its neighbors and their status.

EIGRP Configuration:

Router>enable

Router# configure terminal

Router(config)# router eigrp 100 (AS no.)

Router(config-router)# network 192.168.1.0

Router(config-router)# network 192.168.2.0

Router(config-router)# no auto-summary

Router(config-router)# exit

Router(config)# exit